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Babel's Machines: Linguistic Relativity, Cooperation, and the Emerging Behaviour of Multilingual Artificial Agents

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Large language model (LLM) agents now negotiate, deceive, cooperate, and form coalitions in interactive environments that look more like field studies than benchmarks. A striking empirical signature is emerging across this new behavioural substrate: cooperative tendencies, theory-of-mind performance, and strategic dispositions shift systematically with the language an agent is asked to inhabit — an "Algorithmic Sapir–Whorf Switch" [1] that calls for analysis with the tools of evolutionary game theory, cultural evolution, and statistical physics of cooperation, not engineering benchmarks alone. Taking the machine-behaviour programme of Rahwan et al. [2] as our point of departure, this perspective synthesises three converging lines of inquiry. *First*, we characterise the behavioural phenotype: cross-linguistic divergence in strategy distributions, an unexpected prosocial default that exceeds typical human cooperation rates, and a measurable erosion of theory-of-mind across languages. *Second*, we connect these shifts to mechanistic accounts — the Semantic Hub and Translation Barrier hypotheses — that replace cultural-essentialist explanations with architectural ones. *Third*, we place LLM cooperation in the tradition of Axelrod, Nowak, Boyd, and Perc, arguing that machine cooperation is best understood not as evolved reciprocity but as *culturally inherited cooperation* delivered by pretraining and RLHF, with language acting as the inheritance bottleneck. We foreground contradictory evidence (a cross-lingual persuasion null finding, framing effects that dominate language effects, bimodal Arabic strategy distributions) to bound the claim, and close with four testable predictions for an alignment-relevant science of multilingual machine behaviour.

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1. Introduction: Machine Behaviour Meets Babel

A generation of computational models that once produced text on demand now *acts*. LLM agents negotiate trades, allocate resources, deceive opponents in social-deduction games, and collaborate on open-ended tasks over hours-long horizons [3–6]. They cooperate at rates that exceed typical human players in iterated social dilemmas [7], manage budgets in auction arenas [8], and increasingly inhabit multi-agent ecosystems whose dynamics are no longer fully predictable from individual model capabilities [9, 10]. This transition invites, and arguably requires, a new research programme: the *machine-behaviour* paradigm proposed by Rahwan et al. [2], which treats AI systems as a class of behaving agents whose ethology must be studied empirically, with tools drawn from biology, economics, cognitive science, evolutionary game theory [11], and the statistical physics of cooperation [12], rather than solely from the engineering discipline that built them.

This perspective advances a specific thesis within that programme. Multilingual LLM agents are *linguistically relative actors*: their cooperative tendencies, deceptive capacities, and theory-of-mind performance shift with the language they are asked to inhabit. We call this the *Algorithmic Sapir–Whorf Switch*, following recent empirical work [1]: not the strong Whorfian claim that language determines thought, but the weaker, mechanistically grounded claim that language-specific representations inside pretrained models modulate the policies those models execute in interactive environments. The claim is empirical, not metaphysical, and we devote considerable attention to contradictory evidence and confounds.

The question matters for three reasons that map onto the goals of this journal's call for papers on the science of machine behaviour. First, *empirical characterisation*: we need to describe how LLM agents act and interact under varied linguistic pressure, because deployment is global and multilingual by default. Second, *social and cognitive models*: concepts of trust, cooperation, reciprocity, theory of mind, and cultural evolution — developed over decades to understand humans and other animals — are being silently re-instantiated in machine populations [13–15]. Third, and most consequential, *societal dynamics*: LLMs are already embedded in multilingual institutions — customer service, education, public-sector information access, especially in the Global South — and their language-modulated behaviour will reshape human groups in ways we barely measure.

We organise the perspective around these three goals. Section 2 sketches the interactive substrate in which these behaviours arise. Section 3 develops the linguistic-relativity hypothesis, foregrounding convergent *and* contradictory evidence. Section 4 offers mechanistic explanations (Semantic Hub, Translation Barrier) that replace cultural essentialism with architectural reasoning. Section 5 examines theory of mind across languages. Section 6 places these findings in an evolutionary and cultural-evolution frame. Section 7 turns to societal implications and the research agenda. We close with what we see as the most consequential open questions.

2. An Interactive Substrate

To speak of machine *behaviour*, rather than machine *performance*, we need a substrate richer than single-turn question answering. Recent literature has converged, roughly, on seven families of interactive environments in which LLM agents are now regularly evaluated.

Social-dilemma games (Iterated Prisoner's Dilemma, Public Goods, Ultimatum) provide the mathematically purest view of cooperation, trust, and reciprocity [5, 16, 17]. *Negotiation and diplomacy* (Diplomacy, bargaining games, SPIN-Bench) test persuasion and coalition formation under partial information [3, 18, 19]. *Persuasion and auction* environments probe rhetorical influence and strategic bidding [8, 20]. *Social-deduction games* (Werewolf, Avalon, Spyfall) demand deception and recursive mental-state modelling [21–23]. *Cooperative communication games* (Codenames, Hanabi, Overcooked) require shared cultural and referential ground [24–26]. *Open-world and video-game environments* (Minecraft via Voyager, BALROG, GVGAI-LLM) test long-horizon planning in grounded worlds [4, 27, 28]. *Dialogue-game frameworks* such as clembench provide the most controlled multilingual evaluation, separating format adherence from strategic competence [29–31].

Two observations matter for the argument that follows. First, these environments produce *behaviours*, not just answers: commitments honoured or broken, beliefs propagated or manipulated, coalitions joined or abandoned. Second, every one of these environments has now been run in multiple languages, and every one of them has reported language-dependent divergence — including the "nicer than human" cooperation phenomenon [7, 32] and the substantial English–Arabic performance gap reported by TCG-Bench at large scale [33].

3. The Linguistic-Relativity Hypothesis for Machines

The classical hypothesis of linguistic relativity, associated with Sapir, Whorf, and a century of experimental follow-ups, holds that the language one speaks shapes the cognitive categories one easily forms and deploys [34, 35]. Contemporary research has moved past the strong determinist reading toward a measured view: language provides *cognitive affordances*, subtly biasing attention, memory, and decision-making without dictating them.

We propose that an analogous effect operates in multilingual LLMs. Pretraining on unbalanced, largely English-dominant corpora [36, 37], followed by reinforcement learning from human feedback (RLHF) whose safety and value data are themselves linguistically skewed [38–40], produces models whose internal

representations encode a language-specific inductive bias. When prompted in a different language, the resulting policy shifts.

(a) Cross-Lingual Divergence in Social Dilemmas

High-resolution data from iterated social dilemmas illustrates the phenomenon (Table 1). Across four model families evaluated in five languages, strategy distributions differ meaningfully, but not in ways that map cleanly onto folk-cultural stereotypes [5, 32, 41].

Language	ALLC	ALLD	TFT	WSLS
English	26%	25%	25%	24%
French	27%	18%	26%	29%
Arabic	32%	28%	19%	21%
Chinese	31%	30%	19%	20%
Vietnamese	26%	24%	23%	27%

Table 1. IPD strategy distributions aggregated across four LLM families [5, 32]; LSTM classifier confidence ≥ 0.6. These are model–language interaction effects, not properties of languages.

Three patterns deserve emphasis. Arabic produces *both* the highest unconditional cooperation (32%) *and* the second-highest unconditional defection (28%) — a bimodal, volatile distribution rather than monotonic competitiveness. Vietnamese, often assumed to prime collectivist behaviour, sometimes yields high defection, with cooperative priming collapsing to 2% cooperation in Claude 3.5 Haiku [41]. Chinese favours Win-Stay-Lose-Shift over Tit-for-Tat, an outcome-dependent rather than reciprocity-dependent style. Layered on top, model-family effects are substantial: GPT-4o runs "hawkish" (aggregate ALLD $\approx 31.7\%$) while Mistral runs "dovish" (aggregate ALLC $\approx 33.7\%$) [5].

(b) The "Nicer than Human" Finding

Against these divergences sits a baseline finding that has not received the attention it deserves from the machine-behaviour community. Across multiple recent IPD studies, LLMs default to *prosocial* play at rates substantially exceeding typical human players in comparable settings: Fontana et al. [7] report that Llama2 and GPT-3.5 are "markedly more cooperative than humans" even against random adversaries with substantial defection probabilities, and follow-up work in social dilemmas with multilingual variants documents similar prosocial bias [32]. The picture is genre-dependent, however: in non-cooperative competitive settings such as iterated rock–paper–scissors, frontier LLMs can in fact display *deeper* strategic behaviour than humans [42], suggesting that the prosocial default is specific to cooperative or mixed-motive game structures rather than a uniform property. This is striking evolutionarily: thirty years of agent-based models following Axelrod [13] suggest that stable cooperation requires repeated interactions, reputation, and punishment. LLM cooperation is observed even when these scaffolds are weak, suggesting that cultural inheritance via pretraining produces a default disposition that evolutionary game theory would not predict from interaction structure alone. We return to this in Section 6.

(c) Contradictory Evidence and Confounds

A responsible account of linguistic relativity in machines must foreground the evidence against it. Three findings are central. First, in a controlled cross-lingual study of persuasive co-writing for charitable appeals, Biswas et al. [20] find that prior exposure to a Spanish LLM systematically reduces subsequent use of an English LLM, yet these usage patterns *do not translate into aggregate differences in persuasiveness*; instead, recipients' *beliefs* about whether the advertisement was AI-generated drive donation behaviour, particularly among Spanish-speaking female participants who downwardly adjusted donations when they suspected AI authorship. Second, when game horizons are unknown, cooperation *universalises* across all languages [17, 43]: framing effects can fully dominate language effects. Third, apparent Vietnamese "defection" in low-stakes rounds sometimes reflects arithmetic miscomprehension under attenuated payoffs rather than strategic intent [41].

These findings bound the claim. Language effects exist, but they interact with framing, stakes, and task structure; they do not operate as a fixed cultural dial. Confounds are equally serious: tokeniser quality differs sharply across languages [44], RLHF safety coverage is skewed [38, 45], training corpora are unevenly distributed, and decoding hyperparameters interact with language-specific token statistics. Any causal claim about "language X produces behaviour Y" must be disentangled from these.

4. Mechanism: Semantic Hub and Translation Barrier

Two complementary accounts from mechanistic interpretability provide the machine-behaviour explanation for these patterns, and crucially for reviewers, dispense with the need for cultural-essentialist reasoning.

The *Semantic Hub Hypothesis* [46], supported by logit-lens decoding [47], holds that multilingual LLMs project heterogeneous linguistic inputs into a centralised, English-dominant representational hub in intermediate layers. Causal evidence comes from Activation Addition (ActAdd) experiments [48]: Wu, Yu, et al. [46] show that steering vectors derived in English (e.g., "Good"−"Bad" for sentiment) successfully shift outputs in Spanish and Chinese inputs, often as effectively as native-language steering, indicating that the shared representation space is not a vestigial byproduct but is causally engaged during cross-lingual processing. The corollary — that non-English inputs are routed through English-anchored intermediate

states — predicts behavioural divergences when those intermediate states are sparsely populated for under-represented languages.

The complementary *Translation Barrier Hypothesis* [49] locates many multilingual failures not in upstream reasoning (which proceeds adequately in the hub) but in *late-layer language realisation*: the model reaches the correct conclusion internally and then fails while projecting into the target language's token space. This predicts two diagnostic categories: *hub failures* (intermediate representations do not encode the correct intent) and *realisation failures* (intermediate representations are correct but surface output is not). Language-aware interpretability tools such as Indic-TunedLens offer methodology for the distinction [50].

Together these accounts predict the observed pattern: the bimodal Arabic IPD distribution and Vietnamese arithmetic errors reflect hub-level representational sparsity, while clembench format-adherence failures in German and Italian plausibly reflect late-layer realisation bottlenecks [30, 31]. The same architecture produces both phenomena. A critical corollary is that *confidence calibration* degrades in parallel: Expected Calibration Error rises roughly fivefold from English (4.6%) to non-English (23.1%) prompts [51, 52], producing over-confident, under-reliable agents precisely in the linguistic contexts where their substantive reasoning is weakest.

5. Theory of Mind Across Languages

Cooperation and deception both require *theory of mind* (ToM): the capacity to model the mental states of other agents [53, 54]. In the cognitive-science canon, ToM is often the bottleneck behind cooperative achievement, and its cross-linguistic robustness matters for how LLM agents will perform in human multilingual groups.

The XToM benchmark [55] reveals what its authors describe as a "pronounced dissonance": while frontier models (GPT-4o, DeepSeek-R1) excel at multilingual language understanding, their ToM accuracy varies sharply across languages on the same belief-reasoning items. GPT-4o-11-20, for instance, exhibits an 11-percentage-point gap between Japanese and Chinese on XFANToM belief questions, and most evaluated LLMs — with the notable exception of DeepSeek-R1 — lack robust multilingual ToM reasoning despite achieving comparable accuracy on parallel fact-retrieval items. LLM-Hanabi [25] further establishes that *first-order* ToM (interpreting another agent's intent) is the stronger predictor of cooperative success; when first-order ToM erodes in lower-resource languages, both cooperative *and* deceptive reasoning suffer together. The result is counter-intuitive but coherent: a machine agent with weaker multilingual ToM is simultaneously a less effective collaborator and a less effective deceiver — a combination that is safer on one axis but more brittle on another.

The intent–action gap provides a second cognitive measurement point. The MAST taxonomy, built from over 1,600 execution traces across seven multi-agent frameworks [9], categorises 14 failure modes into Specification/Design Issues (41.77%), Inter-Agent Misalignment (36.94%), and Task Verification (21.30%). The core failures — reasoning-action mismatches and unwarranted conversation resets — are cognitive in nature, and M3-BENCH's *commitment-reliability* metric shows that promise-keeping drops significantly in mixed-motive multilingual games [56].

SPIN-Bench sharpens the picture by decoupling planning from social intelligence [19]: o-series reasoning models (o1, o1-preview, o1-mini) lead formal PDDL classical planning, yet *introducing negotiation phases in Diplomacy degrades* the chain-of-thought coherence of the same strong planners, with o1's deal acceptance rate plateauing around 0.67 and other strong planners performing markedly worse on social metrics. Planning and ToM are distinct faculties that do not scale together. From a machine-behaviour standpoint, this mirrors the evolutionary-psychology distinction between "Machiavellian intelligence" and domain-general problem solving [57], and suggests that cross-lingual transfer may be uneven across these faculties in ways that demand separate measurement.

6. Evolutionary Perspectives on Machine Cooperation

The "nicer than human" finding (Section 3) demands an evolutionary-game-theoretic explanation, and connecting LLM behaviour to this tradition is where the machine-behaviour research programme is least developed and most promising. Two intellectual lineages are immediately relevant. Han [11] has argued that evolutionary game theory provides the natural lens for studying emergent behaviour in multi-agent systems, including agents whose strategies arise from learning rather than mutation; Perc et al. [12] have shown that the statistical physics of cooperation — spatial reciprocity, network reciprocity, phase transitions in cooperative regimes — generalises beyond human populations to any well-defined population of strategic interactors. We propose that multilingual LLM populations satisfy the conditions of both frames.

(a) Axelrod's Legacy in a New Substrate

Axelrod's computer tournaments established that cooperation can emerge among self-interested agents when interactions are iterated, payoffs favour mutual gains, and simple reciprocity (Tit-for-Tat) is available [13, 58]. Subsequent work enriched this with reputation mechanisms, punishment, spatial structure, and indirect reciprocity [59, 60]. In all of these, cooperation is an *achievement*, selected over a history of interactions.

LLM cooperation does not fit this mould. Models arrive at their first interaction *pre-cooperative*: the pretraining and RLHF pipelines act as a form of *cultural inheritance* that delivers a prosocial disposition before any game is played. This is most naturally read as a Boydian–Richersonian cultural-evolution process [15, 61, 62]: pretraining corpora

are cumulative cultural products of millions of human authors, and RLHF injects an additional layer of selected human preferences. The resulting agents exhibit *culturally inherited cooperation*, not evolved reciprocity.

This reframes the linguistic-relativity finding. If cooperation is culturally inherited via text, then languages with different textual corpora should deliver different cooperative priors — exactly the pattern observed in Table 1. The prediction is testable: languages whose pretraining corpora are more heavily weighted toward cooperative, prosocial discourse (educational and informational genres, perhaps) should yield more cooperative play, while corpora dominated by adversarial or commercial text should yield more competitive defaults. Existing corpus-audit methods [63, 64] provide the machinery; the hypothesis has not yet been tested directly.

(b) Cultural Evolution of Machine Behaviour

A further prediction follows. If LLM populations are linked — via shared benchmarks, shared model families, and an increasingly common pipeline of distillation and self-training on model outputs — then machine behaviour is itself subject to cultural evolution. Model-to-model transmission (distillation, synthetic-data training) is a form of *high-fidelity oblique inheritance*; RLHF preferences are a form of *selection*. The long-horizon question is whether machine behavioural traits drift in directions that human selection actively desires, or in directions driven by the structure of the transmission process itself — a classical cultural-evolution question that the machine-behaviour programme is now equipped to ask empirically.

Language interacts with this process. If the transmission bottleneck is largely English — because English is the lingua franca of benchmarks, distillation targets, and RLHF annotator pools — then non-English behavioural traits may drift independently, neither as evolved nor as selected, simply as under-observed side effects. The Semantic Hub Hypothesis supplies the architectural substrate for this prediction [46]. Empirically, the substantial English–Arabic performance gap reported in TCG-Bench at the 32B-parameter scale [33] is consistent with a drift that scale alone does not correct.

(c) Trust, Reciprocity, and Commitment

Four concepts meet in this section. *Trust* and *reciprocity* are operationalised in commitment reliability [56]: when agents announce cooperation and then defect, trust among machine and human counterparts erodes. *Cooperation* is the behavioural output measured in Table 1 and the "nicer than human" finding [7]. *Cultural evolution* [15, 61] is the transmission mechanism producing these behaviours without in-game selection, and *theory of mind* is the cognitive prerequisite that limits how reliably trust and reciprocity can be implemented (Section 5). Framing LLM social behaviour in these terms makes it commensurable with a century of work in biology, anthropology, economics, and the statistical physics of cooperation [12], rather than treating machines as a brand-new object requiring a brand-new vocabulary.

7. Societal Implications and Open Questions

Multilingual LLM agents are not a laboratory curiosity. They are being deployed, now, in institutional contexts where language is the primary mediator of human outcomes: education platforms in Swahili and Hindi, public-sector chatbots in Arabic and Vietnamese, customer-service agents across dozens of markets, and increasingly, autonomous multi-agent systems negotiating commercial transactions [45, 65]. If language modulates machine behaviour as Sections 3–6 suggest, three societal consequences warrant attention.

Unequal quality of machine-mediated institutions. If confidence calibration degrades fivefold outside English [52], public-facing LLM agents in the Global South will systematically offer more over-confident, less reliable advice to populations with fewer alternatives. This is not a bug of individual models but a property of the pipeline.

Linguistically uneven attack surfaces. Code-Switching Red-Teaming achieves a 46.7% increase in attack success by mixing languages at the token level [66]; Agent-in-the-Middle attacks exploit inter-agent channels in multilingual multi-agent systems [67]. Adversaries gain disproportionate leverage in exactly the linguistic contexts where safety data is sparsest. This is a safety externality of multilingual deployment, and the regulatory question — how to incentivise safe deployment behaviour without producing perverse over-regulation — is itself an evolutionary-game-theoretic problem [68].

Co-evolution with human multilingual communities. LLM outputs re-enter human discourse, and thence future pretraining corpora. If the machines' language-modulated behavioural defaults shape the text humans produce — in business email, in customer interactions, in education — then the cultural-inheritance loop we described in Section 6 becomes bidirectional. What this means for institutional norms, linguistic diversity, and trust in machine intermediaries is a research agenda, not a solved problem.

(a) Four Testable Predictions

We close with four predictions formulated to be falsifiable with currently available methods. Each is a wager on the framework above.

P1 — Corpus–cooperation correspondence. Holding model architecture and RLHF pipeline fixed, the rank order of cooperation rates across languages in iterated social dilemmas should track the rank order of cooperative-discourse density in those languages' pretraining corpora, as estimated by existing corpus-audit tools [63, 64]. If

no such correspondence is found, the cultural-inheritance reading of LLM cooperation (Section 6) requires revision.

P2 — Hub–barrier dissociation under intermediate probes. Pre-registered probes at layers $\lfloor L/3 \rfloor$ and $\lfloor 2L/3 \rfloor$ should classify multilingual failures into hub-level (intermediate states fail to encode the correct strategic intent) and realisation-level (intermediate states correct, surface output wrong) categories [50], and the proportions should differ systematically across languages. A null result — failures distributed identically across layers and languages — would weaken the architectural account.

P3 — Cultural-evolution drift under distillation. A controlled chain of distillation experiments (model A trained on synthetic data from model B, evaluated in IPD across languages) should show that behavioural traits propagate with high fidelity in English but degrade in lower-resource languages, reproducing a measurable bottleneck. If degradation does not occur, the inheritance-via-distillation account is wrong.

P4 — Evaluation reliability ceiling. For multilingual LLM-as-Judge [69], panels of ≥ 3 heterogeneous judges combined with self-translation and checklist rubrics should raise Fleiss' κ above 0.5 ("moderate" agreement). If the recovery saturates below this, quantitative cross-lingual machine-behaviour science requires bilingual human adjudication, not just better aggregation.

These predictions are not exhaustive. They are chosen because each, if confirmed, would reshape an existing line of NLP or alignment research; and each, if falsified, would force a clean retreat from a specific claim of this perspective.

8. Conclusions

We have argued that language is an active modulator of behaviour in LLM agents, and that understanding this modulation requires integrating three traditions: *behavioural characterisation* in the style of Rahwan's machine-behaviour programme [2]; *mechanistic interpretability*, which supplies the Semantic Hub and Translation Barrier accounts [46, 49]; and *evolutionary, cultural-evolution, and statistical-physics theories of cooperation* [11–13, 15, 61], which provide the vocabulary of trust, reciprocity, inherited cooperation, and transmission dynamics. The emergent phenomena — a prosocial default that exceeds typical human cooperation rates yet erodes language-dependently, asymmetric theory-of-mind degradation, and compounding safety vulnerabilities via code-switching — are not isolated benchmark artefacts but coordinated signatures of a multilingual machine ethology that is only beginning to be measured.

We have been careful to foreground contradictory evidence: a cross-lingual persuasion null finding, framing effects that dominate language effects, and unresolved confounds. The Algorithmic Sapir–Whorf Switch remains a working hypothesis, not a proven law. The four testable predictions of Section (a) are intended as the next move: each is falsifiable with current methods, and together they convert this perspective from a synthesis into a research agenda. We see this agenda as continuous with the long-running engagement of the life and physical sciences with behaviour, cooperation and evolution — one to which machine populations now offer a new, uncomfortably consequential empirical substrate.

Ethics. This perspective synthesises existing literature and involves no new human-subjects research. Cross-linguistic behavioural divergences reported here reflect model–language interactions under specific training distributions, not properties of languages or of human cultures; readers are urged to communicate findings with this framing to avoid cultural essentialism.

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